HKMA

FINANCIAL RESILIENCE SCOPING ENGINE

Strategic Nexus

HKMA iCAST Hong Kong Framework

Threat-Led Penetration Testing Scoping Document — Nexus v33.0

ENTITY	RESILIENCE INDEX	ENGAGEMENT DURATION	CLASSIFICATION
HKMA iCAST Hong Kong Framework	80%	65 Days	Restricted

Financial Resilience Coverage Index

80

TLPT RESILIENCE INDEX

Financial Resilience Coverage Score

This score reflects the breadth of coverage across the four mandatory Threat-Led Penetration Testing pillars required under DORA Article 26, PRA CBEST, and HKMA iCAST.

TLPT COMPLIANCE PILLAR	COVERAGE STATUS
Threat Intelligence Profiling	VALIDATED
Covert Attack Emulation	VALIDATED
Purple Team Collaboration	VALIDATED
Executive Crisis Validation	GAP — MODULE NOT ENGAGED

Executive Resourcing & Schedule Summary

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The following summarizes the operational timeline and resourcing required to execute the defined Threat-Led Penetration Testing modules in alignment with regulatory mandates.

65
TOTAL ENGAGEMENT DAYS

13
EXECUTION WEEKS

4
TLPT PERSONNEL

3
ACTIVE MODULES

Automated TLPT Resourcing Allocation

ROLE	HEADCOUNT	TLPT RESPONSIBILITY	CLEARANCE
Threat Intel & Red Team Lead	1	Strategic oversight, regulatory coordination, TI report authorship, and final QA gate review.	Full Knowledge
Covert Red Team Operator	3	Technical execution, assumed breach operations, exploit development, and stealth persistence.	Full Knowledge
Purple Team Facilitator	1	Replay exercise facilitation, SOC detection tuning, and SIEM rule validation.	Full Knowledge
Client White Team	TBD	Kill-switch authority, CBS owner coordination, and regulatory notification.	Full Knowledge

1.5 Regulatory Alignment: HKMA iCAST Hong Kong Framework

Narrative: Hong Kong Monetary Authority (HKMA) Cybersecurity Fortification Initiative (CFI) iCAST scoping.

REGULATORY BODY Hong Kong Monetary Authority (HKMA) iCAST (Hong Kong)

ARTICLE/REF	TITLE	REQUIREMENT	SCOPING GUIDANCE
CFI 2.0 – Pillar 1	Cybersecurity Assessment Framework (CAF)	Institutions must conduct a CAF self-assessment to determine their cybersecurity maturity level.	All critical systems and business functions serving retail and institutional clients.
CFI 2.0 – Pillar 2	Intelligence-led Cyber Attack Simulation (iCAST)	Perform controlled, covert, intelligence-led simulated attacks on live production systems.	Threat intelligence-derived simulation targeting CBS with systemic financial impact.
CFI 2.0 – Pillar 3	Cybersecurity Ecosystem Development	Contribution to the Hong Kong financial sector cybersecurity ecosystem and information sharing.	Participation in HKMA Cyber Intelligence Sharing Platform (CISP) post-engagement.

Mandatory Evidence Artifacts:

- CAF Self-Assessment Report
- Threat Intelligence Profile (TI-PHASE Output)
- iCAST Execution Log
- Closure Report & Control Effectiveness Summary
- Remediation Plan

CFI 2.0 Three-Pillar Architecture

The HKMA Cybersecurity Fortification Initiative 2.0 (CFI 2.0) is structured across three pillars: Pillar 1 — Cybersecurity Assessment Framework (CAF) self-assessment; Pillar 2 — Intelligence-led Cyber Attack Simulation Testing (iCAST); and Pillar 3 — Cybersecurity Ecosystem Development including participation in the HKMA Cyber Intelligence Sharing Platform (CISP). All three pillars are expected to be addressed in a complete iCAST engagement.

Assume Breach & SOC Detection Validation

iCAST mandates the 'Assume Breach' methodology to assess the detection and response capabilities of the institution's Security Operations Centre (SOC). The Red Team must simulate an adversary who has already gained initial access and must attempt lateral movement to CBS targets, with the Blue Team's detection speed and accuracy formally measured and documented in the execution log.

CAF Self-Assessment Prerequisite

Before an iCAST engagement, the institution must complete the HKMA CAF self-assessment to determine its cybersecurity maturity level. The CAF assessment score informs the depth and focus of the iCAST scope — institutions scoring in lower maturity bands must address foundational control gaps before a full iCAST can be conducted. CAF assessment evidence must be made available to the iCAST provider at scoping kickoff.

CISP Participation Post-Engagement

Following completion of the iCAST engagement, institutions are expected to contribute anonymized threat intelligence and findings to the HKMA Cyber Intelligence Sharing Platform (CISP) in accordance with Pillar 3 obligations. Threat actor TTPs, attack vectors, and detection gaps identified during the engagement should be formatted for CISP submission to benefit the broader Hong Kong financial ecosystem.

iCAST Notification & Regulatory Reporting

Institutions must notify the HKMA before commencing an iCAST engagement. Upon discovery of any critical or systemic vulnerability during execution, the institution must notify HKMA within 2 hours. Post-engagement, a formal report must be submitted to the HKMA within 30 days of the closure meeting, including the execution log, identified gaps, and a remediation timeline.

ATT&CK Financial Sector Threat Groups

Execution scenarios must reference specific MITRE ATT&CK threat groups active against the financial sector. Key groups include: FIN7 (T1566, T1078, T1071), Carbanak (T1059, T1105), Lazarus Group (T1190, T1133, T1486), and TA505 (T1566.001, T1204). Attack chains must be constructed using the ATT&CK Navigator to demonstrate coverage across Initial Access, Execution, Persistence, Privilege Escalation, and Exfiltration tactics.

Kill Chain Mapping & TTP Evidence

Each attack scenario must produce a formal Kill Chain map documenting the exact sequence of ATT&CK techniques employed. Evidence of each technique must be captured in the form of tool output, command history, and screenshots. The Kill Chain document forms a mandatory annex to the TLPT Final Report and must be provided to the regulator upon request.

IMPLEMENTATION GUIDANCE

OWASP

A01:2021 — Broken Access Control

Testing must validate that users cannot act outside of their intended permissions. For financial institutions, this specifically includes testing of role-based access controls on banking APIs, separation of duties in payment authorization workflows, and privilege escalation paths on core banking administration interfaces.

A02:2021 — Cryptographic Failures

Testing must verify the protection of sensitive data in transit and at rest. For regulated financial data, this includes PAN data (PCI-DSS scope), customer PII, and interbank settlement data. Weak cipher suites, expired TLS certificates, and improperly stored credentials must all be tested.

A10:2021 — Server-Side Request Forgery (SSRF)

SSRF vulnerabilities are particularly dangerous in financial API environments as they can allow an attacker to pivot from a public-facing API to internal payment infrastructure. All public-facing Open Banking APIs must be tested for SSRF to prevent lateral movement from external to internal payment networks.

IMPLEMENTATION GUIDANCE

CIS

Control 1 & 2: Asset & Software Inventory

Actively manage all enterprise assets and software. For TLPT engagements, the CBS Register serves as the asset inventory baseline. Any asset discovered during the Red Team execution that was not in the CBS Register must be flagged as a shadow IT finding and reported as a governance gap.

Control 6: Access Control Management

Testing must validate privileged access management (PAM) controls, MFA enforcement on all privileged accounts, and the absence of default or shared credentials on network devices and servers.

Control 17: Incident Response Management

The Purple Team replay exercises must test the institution's incident response capabilities against the CIS Control 17 framework, measuring time-to-detect, time-to-contain, and time-to-recover against the defined TLPT attack scenarios.

Framework Specific Guidance

HKMA Implementation Guidance

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MITRE Implementation Guidance

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Threat Scenarios & Modeling

PART I.C: THREAT INTELLIGENCE

Threat Intelligence Profiles

> **MANDATORY REQUIREMENT PENDING:** A Threat Intelligence phase or report must be completed and formally appended to define the attack scenarios before Red Team or Intelligence-Led testing begins.

2.1 Digital Chain of Evidence

Artifacts produced during this engagement follow a cryptographically-verifiable lifecycle.

LIFECYCLE PHASE	VALIDATION METHOD
Payload & Exploit Transmission	Zero-Knowledge End-to-End Encryption (E2EE)
Exfiltrated Data Storage	Ephemeral Air-Gapped Analysis Environment (256-bit AES)
Red Team Telemetry Retention	Secure SOC Aggregation (Strict 30-Day Limit)
Artifact Destruction	NIST 800-88 Compliant Cryptographic Wipe

2.2 Data Sovereignty & Residency

To adhere to regulatory data protection requirements, the following residency rules apply to all engagement artifacts.

FRAMEWORK	RESIDENCY MANDATE
HKMA	Assessment data and execution artifacts should be stored locally in Hong Kong in accordance with HKMA guidelines and the Personal Data (Privacy) Ordinance (PDPO).

2.3 Regulator Audit & Access Rights

The following audit rights and access privileges are contractually granted to the respective regulatory bodies.

REGULATOR	ACCESS RIGHTS
HKMA	HKMA retains the right to directly review threat intelligence models, execution logs, and remediation plans. Institutions must notify HKMA of any critical vulnerabilities discovered during iCAST.

Operational Team Governance

PART II.A: TEAM STRUCTURE

STRICTLY CONFIDENTIAL // STRATEGIC SCOPE

White Team & Control Group Structure

In accordance with intelligence-led frameworks (CBEST/iCAST), the engagement must be overseen by a strictly segregated Control Group to ensure operational safety without alerting the defensive Blue Team.

GOVERNANCE BODY	REGULATORY RESPONSIBILITY	CLEARANCE LEVEL
The White Team	Engagement sponsors holding the emergency Kill-Switch. Responsible for defining the Critical Business Services (CBS) and Impact Tolerances.	Full Knowledge
The Control Group	A trusted subset of the client's internal IT/Security team who monitor the engagement for safety but do not participate in defense.	Full Knowledge
The Blue Team	The operational SOC and incident responders who must defend the network as if under a real attack.	Zero Knowledge
Threat Intel Provider	Independent vendor supplying the threat landscape profiles and attack scenarios to the Red Team.	Execution Phase
Covert Red Team	The execution unit simulating the Threat Actor's TTPs against the CBS.	Execution Phase

Critical Business Services & Impact Tolerance

PART II.C: CBS IMPACT TOLERANCE

Critical Business Services (CBS) Scope

⚠ **ACTION REQUIRED — CBS DECLARATION MISSING:** No Critical Business Services have been declared. Under DORA Article 26 and PRA CBEST, the client must map target infrastructure to its underlying business function and formally define the maximum impact tolerance before testing may begin.

Access & Test Data Readiness

> Standard testing prerequisites apply: Client must provide stable network access (VPN/Jump host), valid application credentials (min. 2 per role), and any required API documentation prior to the start of the testing window.

Third-Party ICT Supply Chain Scope

PART II.D: THIRD-PARTY RISK

Third-Party ICT Dependency Register

> No third-party ICT dependencies or hosted environments have been declared in scope. Testing is restricted to first-party infrastructure.

Technical Scope of Work

TI-PHASE — **Targeted Threat Intelligence Gathering**

Formally scoped intelligence gathering phase designed to identify threat actors, define attack scenarios, and produce the Threat Intelligence Profile required before Red Team execution.

Allocated Schedule: 15 Days

RT-COVERT — **Covert Red Team Execution**

Intelligence-led, objective-based penetration testing executing the scenarios defined in the TI-PHASE. Includes stealth, evasion, and assumed-breach scenarios against core financial systems.

Allocated Schedule: 40 Days

PT-REPLAY — **Purple Team Replay Exercises**

Collaborative replay of Red Team attack chains alongside the defensive Blue Team to validate SOC detections, incident response capabilities, and SIEM rule effectiveness.

Allocated Schedule: 10 Days

Execution Timeline (Gantt)

Financial Control Mapping (Omni Matrix)

The following TLPT-specific financial controls are in scope for the selected execution modules, mapped to their regulatory article references.

CONTROL ID	CONTROL NAME	FRAMEWORK	REGULATORY REFERENCE	IMPACT RATING
CB_01	Core Banking Network Segmentation	DORA	Art. 24 (TLPT)	**Critical**
CB_02	Retail Payment Infrastructure Isolation	DORA	Art. 26 (ICT Resilience)	**Critical**
SW_01	SWIFT Gateway Network Isolation	PRA CBEST	Objective A – Segregation	**Critical**
SW_02	SWIFT Messaging Integrity Validation	PRA CBEST	Objective B – Data Integrity	**Critical**
IAM_01	Privileged Access Management (PAM)	FRB	SR 13-1 – Access Control	**High**
IAM_02	Service Account Credential Hygiene	MAS TRM	Section 12.3 – IAM	**High**
IAM_03	Multi-Factor Authentication Enforcement	MAS TRM	Section 12.4 – Authentication	**High**
TI_01	Threat Intelligence Profile Development	HKMA iCAST	CFI 2.0 – TI Framework	**High**
TI_02	Threat Actor Attribution & Profiling	PRA CBEST	CBEST TI Module	**High**
TI_03	Attack Scenario Development (MITRE ATT&CK)	HKMA iCAST	CFI 2.0 – Scenario Modelling	**High**
TI_04	Threat-Led Defense Posture Assessment	HKMA iCAST	CFI-2 – Posture Review	**High**
IR_03	SOC Detection & SIEM Validation	MAS TRM	Section 13 – Incident Response	**High**
IR_04	Purple Team Replay & Remediation Verification	FRB	SR 11-7 – Model Validation	**Medium**
DR_02	RTO / RPO Objective Testing Under Load	FRB	SR 13-1 – Continuity	**High**
API_01	Open Banking API Security Boundary	MAS TRM	Section 12.1 – API Controls	**High**
CLD_01	Cloud Workload Lateral Movement Detection	DORA	Art. 30 – ICT Third-Party Risk	**High**

RACI Responsibility Assignment Matrix

Defines who is Responsible, Accountable, Consulted, and Informed for each TLPT activity across all parties.

ACTIVITY	THREAT INTEL PROVIDER	COVERT RED TEAM LEAD	CLIENT WHITE TEAM	CLIENT CONTROL GROUP	CLIENT SECURITY OPS (SOC)	CLIENT EXEC SPONSOR / CISO
Targeted Threat Intelligence Gathering	R	C	A	I	I	I
Attack Scenario & TTP Development	R	C	A	I	I	I
Critical Business Service (CBS) Declaration	I	I	A	C	C	R
Regulator & Scope Notification	I	C	A	I	I	R
White Team Briefing & Kill-Switch Assignment	I	C	R	C	I	A
Covert Red Team Execution (TLPT)	C	R	A	I	I	I
Emergency Kill-Switch Activation	I	C	R	C	I	A
Purple Team Replay Exercises	C	R	I	I	A	I
SOC Detection Rate Validation	I	C	A	C	R	I
Executive Crisis Management Tabletop	I	C	A	I	I	R
Remediation Planning & Sign-off	I	C	C	I	R	A
Regulator Artifact Submission	C	C	I	I	R	A

Regulatory Deliverables Schedule

Regulatory Deliverables Schedule

The following reports and evidence logs are contractually required to satisfy the loaded regulatory frameworks.

REGULATORY BODY	MANDATORY DELIVERABLE ARTIFACT
Hong Kong Monetary Authority (HKMA) iCAST	CAF Self-Assessment Report
Hong Kong Monetary Authority (HKMA) iCAST	Threat Intelligence Profile (TI-PHASE Output)
Hong Kong Monetary Authority (HKMA) iCAST	iCAST Execution Log
Hong Kong Monetary Authority (HKMA) iCAST	Closure Report & Control Effectiveness Summary
Hong Kong Monetary Authority (HKMA) iCAST	Remediation Plan

Quality Assurance & Peer Review

PART IV.B: QUALITY ASSURANCE

Formal Quality Assurance Gate

Based on the selected regulatory frameworks and execution modules, the following QA routing is algorithmically enforced.

REQUIRED QA LEVEL: TLPT Qualified Red Team Reviewer

Mandatory Threat-Led Penetration Testing (TLPT) review gate. Evidence and kill-chain paths must be validated by a certified Red Team peer reviewer before report release.

Verification & Sign-off

PART V: AUTHORIZATION

NEXUS SEAL

Executive Sponsor Approval

Lead Strategic Provider

Engagement Sign Off & Authorization

By signing below, the parties confirm that this Nexus Scoping Document has been reviewed and is accepted as the binding scope of work for the defined Threat-Led Penetration Testing engagement. This document satisfies the engagement scoping requirements for the active regulatory frameworks: **DORA, PRA CBEST, HKMA iCAST, FRB, and MAS TRM**.

SIGNATORY ROLE	FULL NAME	SIGNATURE	DATE
Client Executive Sponsor / CISO			
Client White Team Lead			
Red Team Lead (Provider)			
Threat Intelligence Lead (Provider)			

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